Dr. JAYASHAKTHI RAJU SARAVANAN., B.D.S., M.D.S., M.S.

jraju.saravanan@gmail.com | +1 (408) 398 2786 | CA, USA

Northeastern University (NEU) & University of Southern California (USC)

Professional Summary

- Health Informatics graduate specializing in predictive analytics, healthcare data security, data analysis, EHR management, and AI-driven solutions.
- Completed 7+ projects in improving healthcare using AI/ML and statistical analysis.
- Passionate in bridging clinical practice with digital health solutions, advancing patient-centered care, data security, optimize workflows, enable data-driven decision-making, and innovation in healthcare.
- 7 years of experience as a clinical consultant. Healthcare professional with a clinical background in periodontology. Published researcher with 6 peer-reviewed articles, and 7 conference presentations.

Educational Qualifications

Master's in health informatics (M.S.) **3.96 GPA** (Jan 2023 - Dec 2024)

Northeastern University, Boston, MA

- Conducted research on leveraging AI in value-based care, utilizing predictive analytics and clinical decision support systems to improve patient outcomes, care coordination, and resource allocation while driving innovation in preventive and value-based healthcare delivery.
- Designed and implemented a relational database system for an online retailer, optimizing workflows, improving data accuracy, and enabling data-driven decision-making for operational scalability.

Master's in public health (M.P.H.) (May 2022 – Aug 2022)

University of Southern California, Keck School of medicine, Los Angeles, CA

- Biostatistics: Applied statistical methods in clinical and public health research using SPSS.
- Health Education and Promotion: Designed health promotion programs in collaboration with the American Red Cross.

Master's in dental surgery (Periodontology) **3.50 GPA** (Aug 2012 - Apr 2015)

J.K.K Nattraja Dental College & Hospital, Komarapalayam, India

- Dissertation: "Estimation of a disintegrin and metalloproteinase 8 levels in gingival crevicular fluid."

Bachelor's in dental surgery **3.72 GPA** (Oct 2006 - Sep 2011)

Sri Ramakrishna Dental College & Hospital, Coimbatore, India

- Participated in multiple rural campaigns to promote awareness about oral health among people.

Technical Skill

- **Technical**: SQL, R, SPSS, Tableau, Amazon Quicksight, Relational Database Management, Data Visualization, Health information standards and interoperability, Generative AI (Google Cloud certified), AI in Healthcare
- **Project Management:** Health IT implementation, reporting, and data standards.
- **Interpersonal:** Strong communication and collaboration with multidisciplinary teams, Critical Thinking, Detail-oriented and Problem Solving.

Professional Certifications

MIT - No Code AI and Machine Learning (Oct 2022 – Dec 2022)

- Theory and applications of AI/ML algorithms using DS tools such RapidMiner, and Ikigai.

Google Cloud – Generative AI in Healthcare (March 2025)

- Explored applications of generative AI in clinical workflows, diagnostics, medical documentation, and personalized care.
- Gained hands-on experience with Google Cloud's AI tools tailored for healthcare use cases (Vertex AI platform).

Professional Experience

- **Experiential network** – The Leap Frog Group, United States (Sep 2024 - Dec2024)
  Developed interactive dashboards with Amazon QuickSight for Leapfrog Group, visualizing healthcare data to support data-driven decision-making and advance quality and safety initiatives.
- **Clinical Consulting Periodontist** – KSJ Dental Clinic, India (Jun 2015 - Mar 2022)
- **Senior Lecturer** – Dept. of Perio., Vinayaka Missions Dental College, India (Jul 2016 - Sep 2016)

Technical Projects

1. **Healthcare Data Analytics & Visualization (*Leapfrog Group*)** Designed interactive dashboards in Amazon QuickSight to visualize hospital performance data, providing real-time insights into hospital ratings, patient safety, and healthcare metrics to support quality and safety improvement initiatives.
2. **Literature Review: AI in Clinical Documentation** Investigated NLP and Deep Learning for automating clinical documentation, researched case studies on DAX Copilot to enhance workflow efficiency, reduce physician burnout, improve data quality, ensure compliance, and optimize healthcare efficiency through automated transcription and predictive analytics.
3. **Leveraging AI in Value-Based Care** *(Health Informatics Capstone Project)* Conducted a literature review on AI in value-based care, evaluating how AI-driven predictive analytics and clinical decision support systems enhance care coordination, cost efficiency, and patient outcomes.
4. **Data Analytics Project: Analyzing NYC Green Taxi Trip Data -** Collaborated on a data analytics project analyzing NYC Green Taxi trip data (2018) using R, **dplyr**, and **ggplot2**. Applied the **CRISP-DM** framework for data preprocessing, handling missing values, and structuring data. Explored trip distributions, fare trends, and tipping behaviors through statistical insights and visualizations. Developed a **K-Nearest Neighbors (KNN)** model, optimizing hyperparameters and evaluating performance to classify trip patterns efficiently.
5. **GIS Mapping for Cardiovascular Disease Risk Analysis (***Data Analytics & Public Health Project*) Developed a data-driven resource allocation approach for high-risk regions by conducting geospatial analysis using GIS to map CVD prevalence among ethnic groups, supporting targeted healthcare interventions.
6. **ICD-10 & Transition to ICD-11 for Improved Clinical Coding** *(Health Informatics Research Project)* Evaluated the impact of ICD-10 on public health surveillance and clinical decision-making, analyzed the transition to ICD-11 for improved semantic interoperability, and examined healthcare data standards in fraud detection, reimbursement, and care optimization.
7. **E-Commerce Data Analytics** Sports Retail Database *(Database Design & SQL Query Optimization Project)* Designed and implemented a relational database for an online sporting goods retailer, conducting SQL-based analysis to optimize inventory management, enhance customer retention, and develop data-driven pricing strategies based on purchasing behavior and demographics.
8. **Hospital Nurse Intervention Pilot Program** Utilized R programming and SQL to analyze ICU and SICU bed capacities for a nursing intervention pilot, developing data-driven reports to identify high-volume hospitals and recommending optimal sites based on bed capacity and staffing needs to enhance patient outcomes.